

Triple DES (3DES)

1. Aim

To implement the Triple DES (3DES) encryption technique using three independent keys applied in sequence, and to understand how applying encryption three times enhances the strength of a symmetric cipher.

2. Algorithm

1. Start the program.
2. Declare variables for plaintext, the three keys k1, k2, k3 and cipher.
3. Read the plaintext value entered by the user.
4. Read the three keys k1, k2 and k3 entered by the user.
5. Encrypt the plaintext by successively performing XOR with k1, then k2, then k3.
6. Display the encrypted value.
7. Decrypt the cipher by performing XOR in the reverse order: first k3, then k2, then k1.
8. Display the decrypted value and stop the program.

3. Program

```
#include <stdio.h>

int main() {
    int plaintext, k1, k2, k3, cipher;

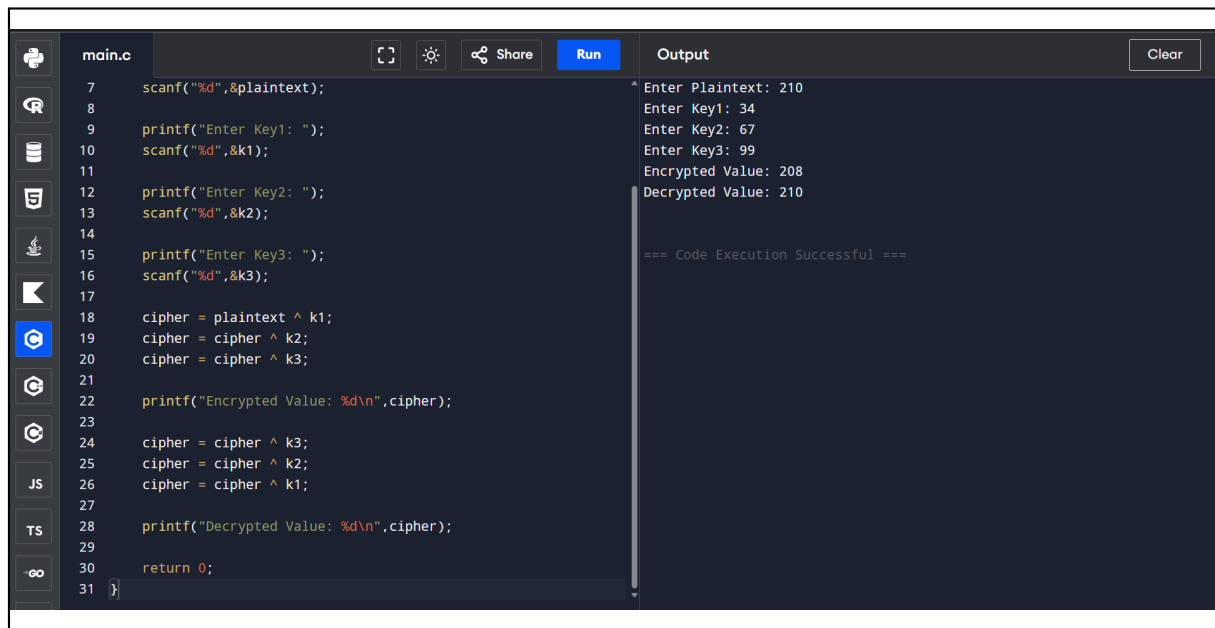
    printf("Enter Plaintext: ");
    scanf("%d", &plaintext);
    printf("Enter Key1: ");
    scanf("%d", &k1);
    printf("Enter Key2: ");
    scanf("%d", &k2);
    printf("Enter Key3: ");
    scanf("%d", &k3);

    cipher = plaintext ^ k1;
    cipher = cipher ^ k2;
    cipher = cipher ^ k3;
    printf("Encrypted Value: %d\n", cipher);

    cipher = cipher ^ k3;
    cipher = cipher ^ k2;
    cipher = cipher ^ k1;
    printf("Decrypted Value: %d\n", cipher);

    return 0;
}
```

4. Output Screenshot



The screenshot shows a C program named `main.c` being executed. The program prompts the user to enter a plaintext value and three keys. It then performs three successive XOR operations to encrypt the plaintext and three successive XOR operations to decrypt the ciphertext back to the original plaintext. The output shows the plaintext as 210, the keys as 34, 67, and 99, the encrypted value as 208, and the decrypted value as 210. The execution is successful.

```
main.c
7  scanf("%d",&plaintext);
8
9  printf("Enter Key1: ");
10 scanf("%d",&k1);
11
12 printf("Enter Key2: ");
13 scanf("%d",&k2);
14
15 printf("Enter Key3: ");
16 scanf("%d",&k3);
17
18 cipher = plaintext ^ k1;
19 cipher = cipher ^ k2;
20 cipher = cipher ^ k3;
21
22 printf("Encrypted Value: %d\n",cipher);
23
24 cipher = cipher ^ k3;
25 cipher = cipher ^ k2;
26 cipher = cipher ^ k1;
27
28 printf("Decrypted Value: %d\n",cipher);
29
30 return 0;
31 }
```

Output

```
Enter Plaintext: 210
Enter Key1: 34
Enter Key2: 67
Enter Key3: 99
Encrypted Value: 208
Decrypted Value: 210

=== Code Execution Successful ===
```

5. Sample Input / Output

```
Enter Plaintext: 120
Enter Key1: 11
Enter Key2: 22
Enter Key3: 33
Encrypted Value: 68
Decrypted Value: 120
```

6. Result

The Triple DES (3DES) program applying three successive XOR key operations was implemented and executed successfully, and the original plaintext was correctly recovered after decryption.